

ReStock AI

A Human-Governed Autonomous Purchasing Officer

The AI handles the volume. The human owns the judgment.

Agentika 2.0 · NaSCon 2026

Human + AI: Better Together

Team Chat GPTwo · Chaudhary Muhammad Huzaifa Ali · Umamah Nasir Khan

The problem

Thousands of restock decisions every week — and both naive answers fail

Full automation

- One mispriced or oversized order silently burns the budget.
- No human sees the risky calls before money is spent.
- Errors scale as fast as the automation does.

Fully manual

- The team drowns in routine approvals.
- Reacting to demand changes is too slow.
- Judgment is wasted on orders that need none.

The routine volume needs automation; the risky calls need human judgment.

The solution: one agent, three routes

A Q-Learning agent proposes; a dual safety gate governs; the human decides the exceptions.

1

Auto-execute

- Routine, confident, affordable orders
- Run automatically — stock replenished
- A real purchase order is issued

2

Escalate

- Expensive / low-confidence / unfamiliar
- LLM writes a plain-English rationale
- Approve or reject — even from Discord

3

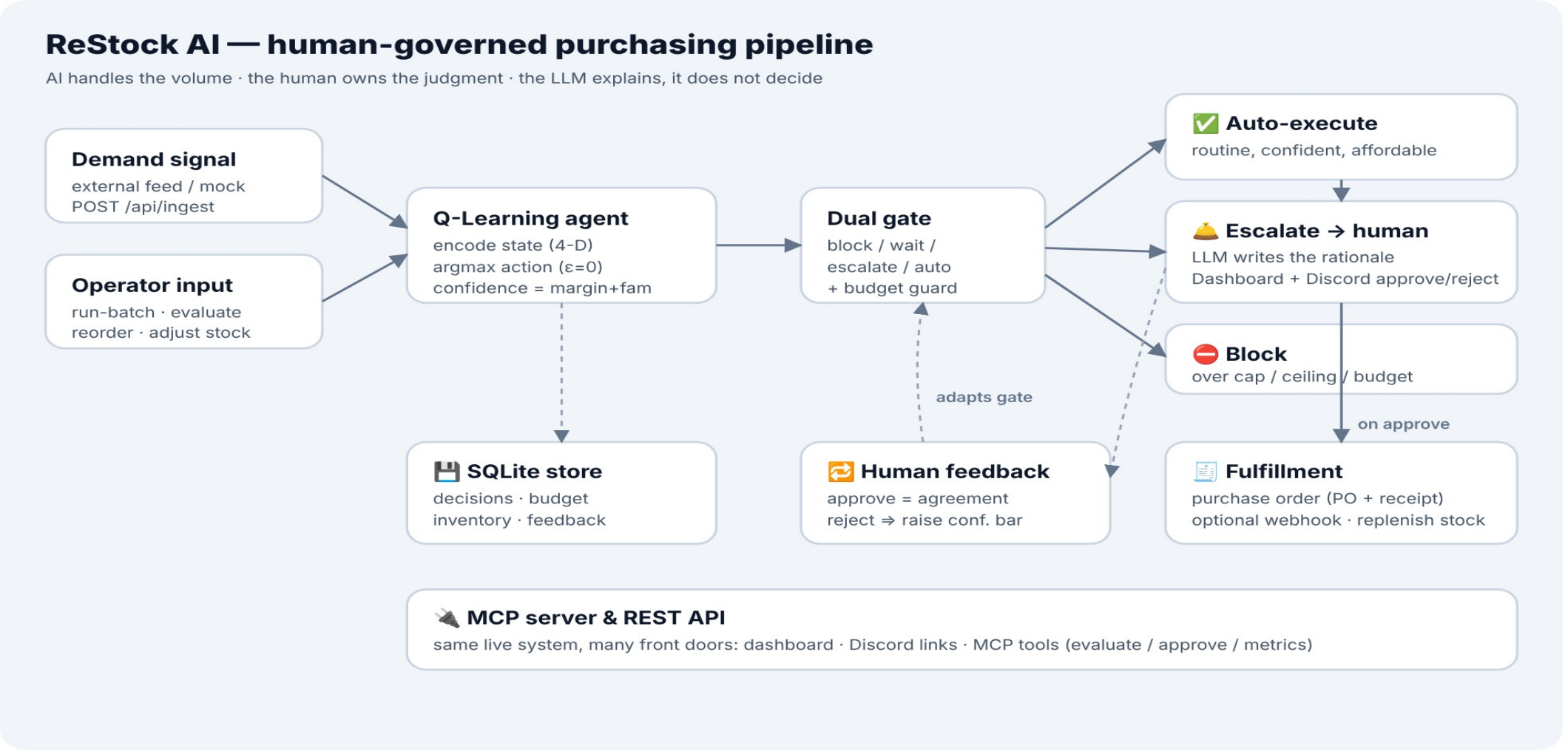
Block

- Over quantity cap, cost ceiling or budget
- Refused outright — no spend
- Recorded with its reasons

The LLM explains — it never decides.

How it works — architecture

One FastAPI service, a Q-Learning core, and four integrations wired around it



The decision flow, end to end

Every situation runs through the same pipeline

1

Encode state

Situation → 4-D state (LOW/MED/HIGH), 81 states

2

Decide

Q-table argmax (ε=0) → action, quantity, confidence

3

Dual gate

Block · Wait · Escalate · Auto — first match wins

4

Act

Execute · escalate to human · or block

5

Fulfill

Issue a PO (+ optional webhook); replenish stock

Confidence = 0.7 × margin + 0.3 × familiarity — margin = how decisively the top action won; familiarity = how well-trained the state is.

The dual gate

First category that matches wins — all applicable reasons are recorded

1	**Block**	qty > MAX_ORDER_QTY, cost > HARD_LIMIT, or cost > budget
2	**Auto-approve**	WAIT actions — a safe no-op
3	**Escalate**	cost > AUTO_LIMIT
4	**Escalate**	confidence < CONF_THRESHOLD (effective)
5	**Escalate**	visits < MIN_VISITS_FAMILIAR (unfamiliar)
6	**Auto-approve**	otherwise — routine, confident, affordable

Human-in-the-loop & the adaptive gate

Escalations wait for a person — on the dashboard or in Discord

Approve

Order executes, stock replenishes, a PO is issued. Recorded as agreement with the AI.

Reject

Order is voided. Recorded as an override of the AI.

Discord sign-off

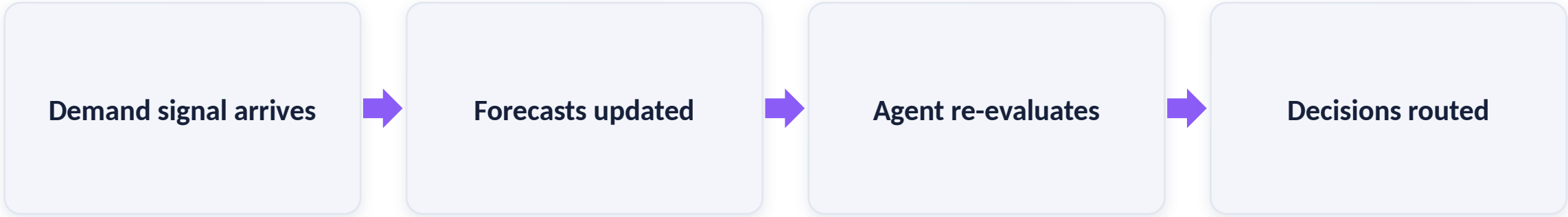
Each escalation is pushed to Discord with one-click Approve / Reject links — decide from your phone.

Adaptive gate

A run of rejections raises the confidence bar, so the gate escalates sooner. Governance only — the model is never retrained.

Reacting to the real world

POST /api/ingest closes the loop from the outside



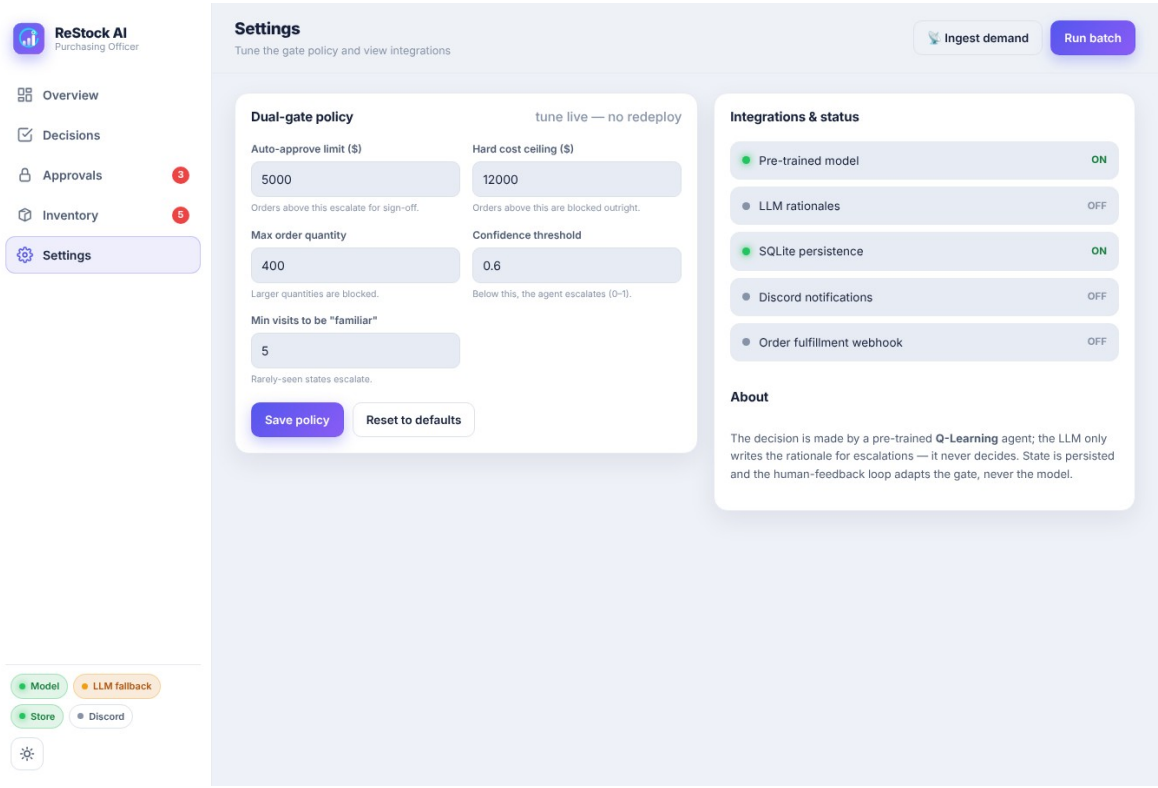
New demand in, decisions out — no human needed for the routine part. The feed can be an external endpoint or a built-in mock, and it degrades gracefully.

A real app — not a dashboard

Sidebar navigation · five pages · light & dark themes · live counters



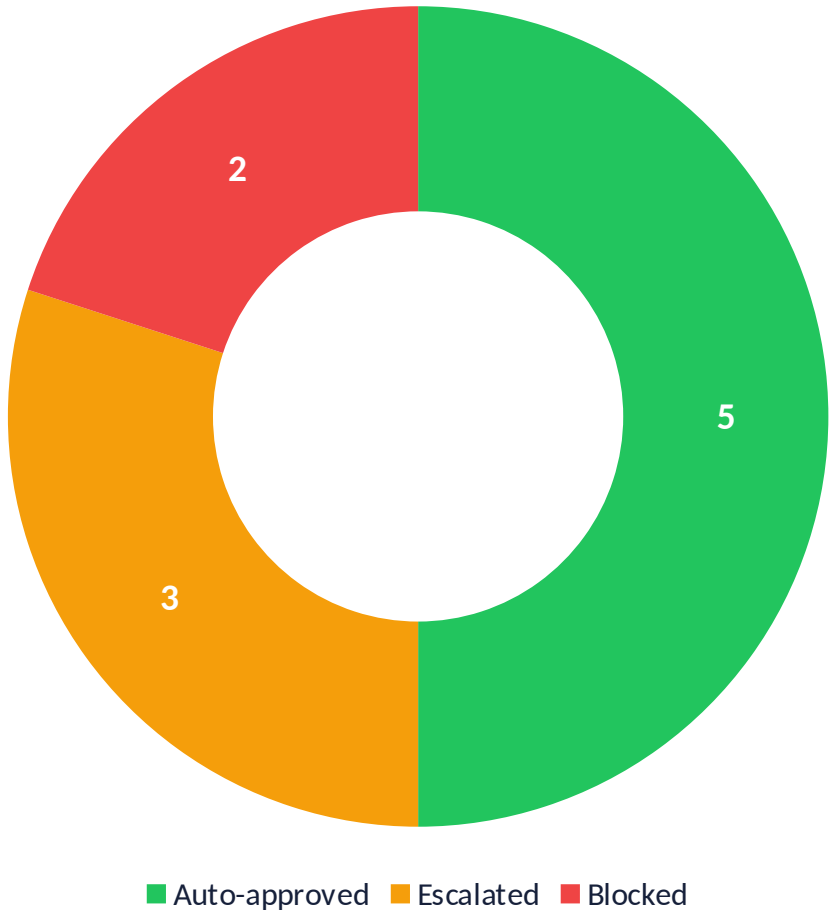
Overview — analytics & routing



Settings — live-edit the gate policy

Decision routing at a glance

A typical batch over the seed catalogue



50% Auto-approved without a human

3 Escalated for human sign-off

2 Blocked — budget protected

100% Decisions logged & auditable

Integrations & technology

Automation depth, coherently wired — all optional, all graceful

Discord / Slack

Approve or reject escalations from chat

Fulfillment webhook

PO receipts POSTed to a downstream system

MCP server

Drive the live service from any MCP client

SQLite persistence

State survives a redeploy

Demand ingestion

External feed or mock → reactive loop

Tested · 24 passing

Gate, confidence, persistence, API

Python · FastAPI · Q-Learning · SQLite · Docker · deployed on Clouds by Alternate Futures

How it meets the judging rubric

100 points across six criteria

Technical execution & reliability	25	End-to-end, graceful degradation, 24 tests
Usefulness & real-world impact	20	Replenishment, budget control, real POs
Theme alignment	15	AI volume + human sign-off on risk
Creativity & uniqueness	15	Dual gate, adaptive gate, reactive loop
Automation depth & integration	15	Discord + webhook + MCP + ingestion
User experience & presentation	10	Full app UI, light/dark, charts

Honest, deployed, and built to the theme

- The Q-table is a synthesised, explainable policy — a drop-in artifact, not a black box.
- The feedback loop adapts the gate, never the model.
- The LLM explains; the Q-table decides.

Live on Alternate Clouds

alternate-restless-forest-3113-github-app.alternatefutures.ai

github.com/chhuzaifa928/Restocker

Thank you — ReStock AI · Team Chat GPTwo